

MATHEMATICAL ANALYSIS / ADDITIVE COMBINATORICS / ACHIEVEMENT SETS

# FINITE CERTIFICATES

## FOR THREE UNRESOLVED CARDINAL-FUNCTION RANGES

Exact finite witnesses realizing  
 $\{1, 2, 3, 4, 5\}$ ,  $\{1, 2, 3, 5, 6\}$ , and  $\{1, 3, 4, 5\}$

### FINITE WITNESSES

$$\begin{aligned} a &= (1, 2, 2, 3, 3), & b &= (1, 1, 2, 2, 3, 6), \\ c &= (2, 5, 6, 9, 10, 11, 13) \end{aligned}$$



# Finite Certificates for Three Unresolved Cardinal-Function Ranges

Exact subset-sum multiplicity spectra for three bounded classification cases

## Metriq PRISM Laboratory

Program for Research in Intelligent Systems and Methods

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### Research status

This manuscript supplies three finite, exact, independently checkable certificates for entries printed as unknown in the August 2025 summary table of Marchwicki and Zmija. Version 2.0 consolidated the seven-atom certificate previously circulated in two superseded draft manuscript files; Version 2.2 is the Metriq PRISM Laboratory edition. The arguments and novelty claims have not yet undergone independent specialist review or peer review.

### Abstract

For a finite sequence  $x = (x_1, \dots, x_N)$  of positive integers, let  $f_x(t)$  count the subsets of indices whose  $x$ -sum is  $t$ . A 2025 classification table leaves several bounded ranges unresolved for finite and Cantor achievement sets. We give the explicit witnesses

$$a = (1, 2, 2, 3, 3), \quad b = (1, 1, 2, 2, 3, 6), \quad c = (2, 5, 6, 9, 10, 11, 13).$$

Their subset-sum generating polynomials have positive coefficient ranges

$$\text{rng}(f_a) = \{1, 2, 3, 4, 5\}, \quad \text{rng}(f_b) = \{1, 2, 3, 5, 6\}, \quad \text{rng}(f_c) = \{1, 3, 4, 5\}.$$

The third certificate is structurally notable because multiplicity 2 is absent while 3, 4, and 5 all occur. Each coefficient identity is verified by exact polynomial multiplication and exhaustive subset enumeration. We then prove one general Cantorization theorem: appending the uniquely represented tail  $(3^{-n})_{n \geq 1}$  preserves every representation count and produces an achievement set homeomorphic to the Cantor set. Under the source table's conventions, the three certificates supply eight affirmative entries previously left blank in the finite, Cantor, and unrestricted classes. The corresponding Cantorval questions remain unresolved.

**Candidate main conclusion.** The finite witnesses  $a$ ,  $b$ , and  $c$  realize the cardinal-function ranges  $\{1, 2, 3, 4, 5\}$ ,  $\{1, 2, 3, 5, 6\}$ , and  $\{1, 3, 4, 5\}$ , respectively. Each admits an explicit Cantor-set extension with the same range.

**MSC 2020.** Primary 40A05; Secondary 05A15, 11B13, 11K31.

**Keywords.** Achievement set, cardinal

function, subset sum, generating polynomial, Cantor set.

## 1 Problem context

Let  $x = (x_n)$  be a finite sequence, or an absolutely summable sequence, of positive real numbers. Its *achievement set* is

$$\mathcal{A}(x) = \left\{ \sum_{n \in S} x_n : S \text{ is a subset of the index set} \right\}.$$

The associated *cardinal function* is

$$f_x(t) = \# \left\{ S : \sum_{n \in S} x_n = t \right\}, \quad t \in \mathcal{A}(x),$$

and its range is  $\text{rng}(f_x) = \{f_x(t) : t \in \mathcal{A}(x)\}$ . Equivalently, if  $P_x(z) = \prod_{i=1}^N (1 + z^{x_i})$ , then  $\text{rng}(f_x)$  is the set of positive coefficients of  $P_x$ ; zero coefficients at unattained integers are not included.

Marchwicki and Żmija classify which bounded sets can occur as such ranges for several topological types of achievement set [1]. Their summary uses:

- $\mathcal{F}$  for ranges realized by finite achievement sets;
- $\mathcal{C}$  for ranges realized by achievement sets homeomorphic to the Cantor set;
- $\mathcal{C}_v$  for ranges realized by Cantorvals;
- $\mathcal{R}$  for all ranges realized by summable positive sequences.

The authors state that blank table entries are unknown. The entries addressed here concern

$$M_5 = \{1, 2, 3, 4, 5\}, \quad M_{356} = \{1, 2, 3, 5, 6\}, \quad M_{1345} = \{1, 3, 4, 5\}.$$

For  $M_5$ , the finite and Cantor entries are blank. For each of  $M_{356}$  and  $M_{1345}$ , the finite, Cantor, and unrestricted entries are blank. The corresponding Cantorval entries are not resolved here.

Table 1: The entries addressed by this manuscript. “Blank” reports the status printed in the August 2025 summary table; “yes” is the candidate update proved below.

| Range      | 2025 table    |               |               | Candidate update |               |               |
|------------|---------------|---------------|---------------|------------------|---------------|---------------|
|            | $\mathcal{F}$ | $\mathcal{C}$ | $\mathcal{R}$ | $\mathcal{F}$    | $\mathcal{C}$ | $\mathcal{R}$ |
| $M_5$      | blank         | blank         | yes           | yes              | yes           | yes           |
| $M_{356}$  | blank         | blank         | blank         | yes              | yes           | yes           |
| $M_{1345}$ | blank         | blank         | blank         | yes              | yes           | yes           |

The arXiv record inspected on 18 July 2026 lists only version 1, submitted 18 August 2025. Targeted searches for the three exact ranges and the three witnesses did not identify a later publication filling these entries. That search cannot establish priority; it only motivates publication of an auditable candidate note.

## 2 Generating-polynomial certificate

The finite problem is equivalent to reading coefficients of a polynomial.

**Lemma 2.1 (Subset-sum coefficient identity).** *Let  $x = (x_1, \dots, x_N)$  be a finite sequence of positive integers. Then*

$$P_x(z) := \prod_{i=1}^N (1 + z^{x_i}) = \sum_{t=0}^{x_1 + \dots + x_N} f_x(t) z^t,$$

where  $f_x(t) = 0$  for unattainable  $t$ .

*Proof.* When the product is expanded, choosing 1 from the  $i$ th factor records that index  $i$  is omitted, while choosing  $z^{x_i}$  records that it is included. Every subset  $S \subseteq \{1, \dots, N\}$  contributes the monomial  $z^{\sum_{i \in S} x_i}$ . The coefficient of  $z^t$  is therefore exactly the number of subsets with sum  $t$ .  $\square$

**Remark 2.2** (Complement symmetry). If  $T = \sum_i x_i$ , subset complementation gives  $f_x(t) = f_x(T - t)$ . Accordingly,  $P_x$  is palindromic. This provides an immediate internal check on both expansions below.

### 3 First certificate: the complete range through five

Set

$$a = (1, 2, 2, 3, 3).$$

**Theorem 3.1.** *The cardinal function of  $a$  has range*

$$\text{rng}(f_a) = \{1, 2, 3, 4, 5\}.$$

Consequently  $M_5 \in \mathcal{F}$ .

*Proof.* By lemma 2.1,

$$\begin{aligned} P_a(z) &= (1+z)(1+z^2)^2(1+z^3)^2 \\ &= 1 + z + 2z^2 + 4z^3 + 3z^4 + 5z^5 + 5z^6 \\ &\quad + 3z^7 + 4z^8 + 2z^9 + z^{10} + z^{11}. \end{aligned}$$

Every coefficient is positive, so  $\mathcal{A}(a) = \{0, 1, \dots, 11\}$ . The set of coefficients is exactly  $\{1, 2, 3, 4, 5\}$ , proving the claim.  $\square$

Table 2: Exact coefficient certificate for  $a = (1, 2, 2, 3, 3)$ .

| $t$      | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|----------|---|---|---|---|---|---|---|---|---|---|----|----|
| $f_a(t)$ | 1 | 1 | 2 | 4 | 3 | 5 | 5 | 3 | 4 | 2 | 1  | 1  |

The total of the coefficients is  $32 = 2^5$ , as required by the number of subsets, and the coefficient sequence is symmetric about  $11/2$ , as required by remark 2.2.

### 4 Second certificate: multiplicity four is omitted

Set

$$b = (1, 1, 2, 2, 3, 6).$$

**Theorem 4.1.** *The cardinal function of  $b$  has range*

$$\text{rng}(f_b) = \{1, 2, 3, 5, 6\}.$$

Consequently  $M_{356} \in \mathcal{F} \subset \mathcal{R}$ .

*Proof.* Again by lemma 2.1,

$$\begin{aligned} P_b(z) &= (1+z)^2(1+z^2)^2(1+z^3)(1+z^6) \\ &= 1 + 2z + 3z^2 + 5z^3 + 5z^4 + 5z^5 + 6z^6 + 5z^7 \\ &\quad + 5z^8 + 6z^9 + 5z^{10} + 5z^{11} + 5z^{12} \\ &\quad + 3z^{13} + 2z^{14} + z^{15}. \end{aligned}$$

Thus every integer from 0 through 15 is attained. The positive coefficients assume precisely the five values 1, 2, 3, 5, 6; the value 4 never occurs.  $\square$

Table 3: Exact coefficient certificate for  $b = (1, 1, 2, 2, 3, 6)$ .

| $t$      | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|----------|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| $f_b(t)$ | 1 | 2 | 3 | 5 | 5 | 5 | 6 | 5 | 5 | 6 | 5  | 5  | 5  | 3  | 2  | 1  |

Here the coefficient total is  $64 = 2^6$ , and the coefficient sequence is palindromic about  $15/2$ .

## 5 Third certificate: multiplicity two is omitted

Set

$$c = (2, 5, 6, 9, 10, 11, 13).$$

**Theorem 5.1 (Seven-atom certificate).** *The cardinal function of  $c$  has range*

$$\text{rng}(f_c) = \{1, 3, 4, 5\}.$$

*In particular, no point of  $\mathcal{A}(c)$  has exactly two representations.*

*Proof.* By lemma 2.1,

$$P_c(z) = (1 + z^2)(1 + z^5)(1 + z^6)(1 + z^9)(1 + z^{10})(1 + z^{11})(1 + z^{13}).$$

Direct multiplication gives the grouped identity

$$P_c(z) = \sum_{t \in S_1} z^t + 3 \sum_{t \in S_3} z^t + 4 \sum_{t \in S_4} z^t + 5 \sum_{t \in S_5} z^t,$$

where

$$\begin{aligned} S_1 &= \{0, 2, 5, 6, 7, 8, 9, 10, 12, 14, 42, 44, 46, 47, 48, 49, 50, 51, 54, 56\}, \\ S_3 &= \{11, 13, 15, 16, 17, 18, 19, 20, 23, 25, 31, 33, 36, 37, 38, 39, 40, 41, 43, 45\}, \\ S_4 &= \{21, 22, 27, 28, 29, 34, 35\}, \\ S_5 &= \{24, 26, 30, 32\}. \end{aligned}$$

The six remaining exponents in  $\{0, 1, \dots, 56\}$  are 1, 3, 4, 52, 53, 55 and have coefficient zero. Thus every positive coefficient is one of 1, 3, 4, 5, and all four values occur. No coefficient is 2.

As checks, the coefficient vector is palindromic about  $t = 28$ , and

$$20 \cdot 1 + 20 \cdot 3 + 7 \cdot 4 + 4 \cdot 5 = 128 = 2^7.$$

Representative fibers are

$$\begin{aligned} 11 &= 11 = 5 + 6 = 2 + 9, \\ 21 &= 10 + 11 = 5 + 6 + 10 = 2 + 9 + 10 = 2 + 6 + 13, \\ 24 &= 11 + 13 = 5 + 9 + 10 = 5 + 6 + 13 = 2 + 9 + 13 = 2 + 5 + 6 + 11. \end{aligned}$$

The complete grouped identity proves globally that multiplicity two never occurs. □

## Exact subset-sum multiplicity spectra

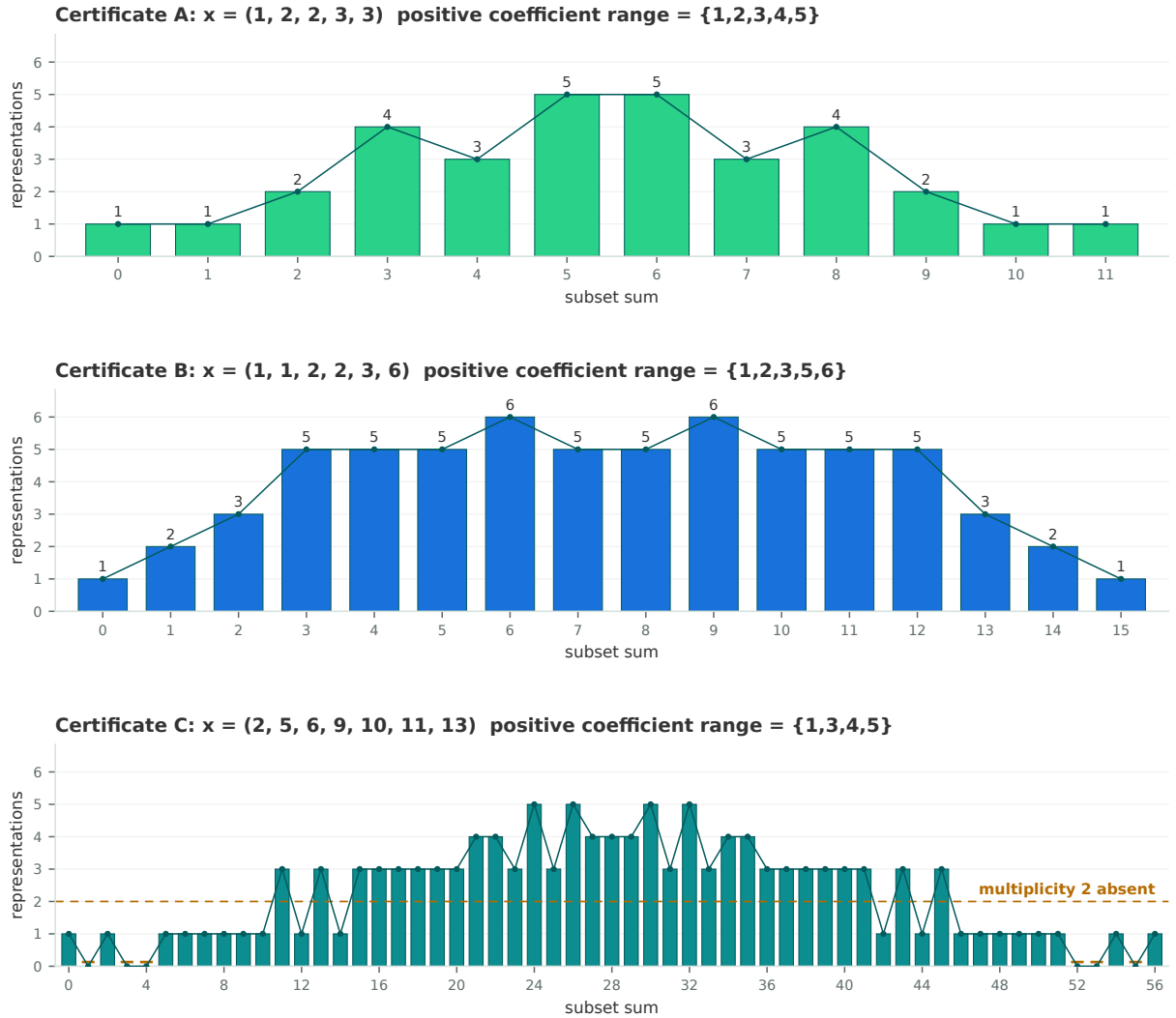


Figure 1: The three exact generating-polynomial coefficient spectra. Certificate A realizes every multiplicity from one through five; Certificate B reaches six while omitting four; Certificate C realizes 1, 3, 4, 5 while omitting two.

## 6 Explicit Cantor realizations with the same ranges

All three finite certificates can be converted directly into Cantor achievement sets without changing any representation count.

Let

$$\tau = (3^{-1}, 3^{-2}, 3^{-3}, \dots)$$

and write

$$K = \mathcal{A}(\tau) = \left\{ \sum_{n=1}^{\infty} \frac{\varepsilon_n}{3^n} : \varepsilon_n \in \{0, 1\} \right\}.$$

Then  $K$  is a one-half scaling of the standard ternary Cantor set and is contained in  $[0, 1/2]$ .

**Lemma 6.1.** *Every point of  $K$  has a unique representation by the sequence  $\tau$ .*

*Proof.* Suppose two binary digit sequences first differ at index  $n$ . The difference contributed by the  $n$ th term has absolute value  $3^{-n}$ , while the largest possible absolute contribution of all later terms is

$$\sum_{k>n} 3^{-k} = \frac{1}{2 \cdot 3^n} < \frac{1}{3^n}.$$

The later terms cannot cancel the first difference. Hence the representations are distinct only when the digit sequences are distinct.  $\square$

For a finite integer sequence  $x$ , let  $x \sqcup \tau$  denote concatenation with the tail  $\tau$ .

**Proposition 6.2 (Cantorization lemma).** *If  $x$  is a finite sequence of positive integers, then*

$$\text{rng}(f_{x \sqcup \tau}) = \text{rng}(f_x),$$

*and  $\mathcal{A}(x \sqcup \tau)$  is a Cantor set.*

*Proof.* Every point of the extended achievement set lies in

$$\mathcal{A}(x \sqcup \tau) = \bigcup_{s \in \mathcal{A}(x)} (s + K).$$

Distinct values  $s, s' \in \mathcal{A}(x)$  are distinct integers and therefore satisfy  $|s - s'| \geq 1$ . Since  $K \subset [0, 1/2]$ , the translates  $s + K$  and  $s' + K$  are disjoint. Thus a point  $t \in s + K$  determines its finite-prefix sum  $s$  uniquely. By lemma 6.1, it also determines its tail digits uniquely. The only freedom is the choice of a subset of the finite prefix summing to  $s$ , so

$$f_{x \sqcup \tau}(t) = f_x(s).$$

This proves equality of the ranges.

Finally,  $\mathcal{A}(x \sqcup \tau)$  is a finite disjoint union of translates of  $K$ . It is nonempty, compact, perfect, and nowhere dense; hence it is homeomorphic to the Cantor set.  $\square$

**Corollary 6.3.** *Under the class notation of Marchwicki and Zmija,*

$$\boxed{\{1, 2, 3, 4, 5\} \in \mathcal{F} \cap \mathcal{C}}, \quad \boxed{\{1, 2, 3, 5, 6\} \in \mathcal{F} \cap \mathcal{C} \subset \mathcal{R}},$$

and

$$\boxed{\{1, 3, 4, 5\} \in \mathcal{F} \cap \mathcal{C} \subset \mathcal{R}}.$$

Thus the constructions fill the eight entries identified in Table 1.

*Proof.* Apply proposition 6.2 to  $a$ ,  $b$ , and  $c$ , using theorems 3.1, 4.1 and 5.1.  $\square$

## Cantor extension preserves every finite-prefix multiplicity

The uniquely represented tail lies in  $[0, 1/2]$ , so translates at distinct integer prefix sums are disjoint.

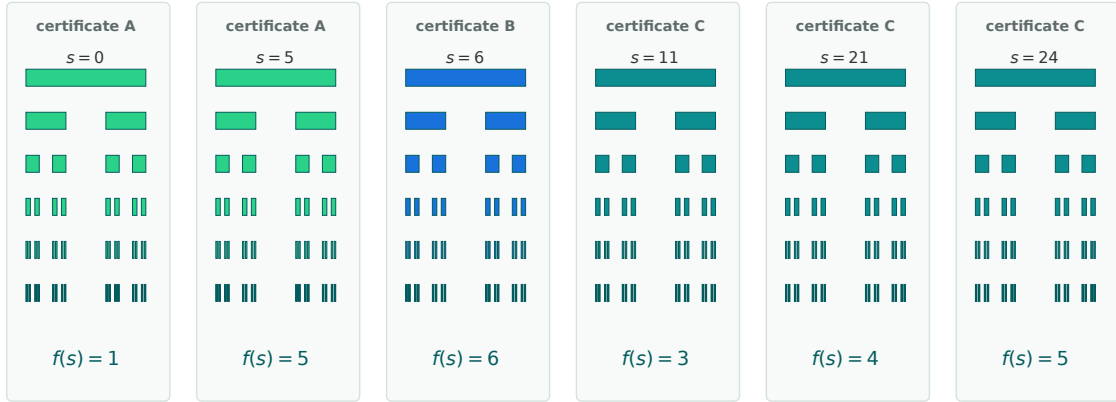


Figure 2: The Cantorization mechanism. The uniquely represented tail has diameter  $1/2$ , so its translates at distinct integer prefix sums are disjoint. Each component therefore inherits exactly the multiplicity of its finite prefix sum.

## 7 Verification and discovery method

The proof is a finite symbolic certificate. It does not depend on numerical approximation, limiting estimates, or probabilistic computation. The accompanying verification script performs two independent exact calculations for each of the three witnesses:

1. dynamic multiplication of the factors  $(1 + z^{x_i})$ ;
2. direct enumeration of all  $2^N$  subsets.

It then checks coefficient equality, coefficient range, complement symmetry, positivity of every displayed coefficient, and the identity  $\sum_t f_x(t) = 2^N$ . Rational arithmetic verifies the tail identities in [lemma 6.1](#) and [proposition 6.2](#).

The witnesses were found during bounded computer-assisted searches over short nondecreasing integer sequences, with polynomial coefficients updated exactly. The search was used only for discovery. Once the sequences are stated, [theorems 3.1](#), [4.1](#) and [5.1](#) are complete hand-checkable proofs. No claim of minimality is made.

### Independent review target

A reviewer can verify the first two certificates by multiplying short factorizations and the third by multiplying seven binomials or enumerating 128 subsets. The only topological extension step is the disjoint-translate argument in [proposition 6.2](#). The principal remaining research questions are novelty, prior art, and whether the cited table is intended exactly as printed.

## 8 Version 2.2 revision and supersession

The canonical release is MF-PRISM-MATH-2026-06, Version 2.2. Its mathematical content is the consolidated three-certificate paper introduced in Version 2.0. The seven-atom certificate was previously circulated in two draft manuscript files; those drafts contain the same witness and theorem and are superseded by this manuscript. They should be archived as superseded drafts rather than cited as separate mathematical contributions. Version 2.2 updates the publisher imprint, cover identity, masthead, citation metadata, and



institutional disclaimer to the Metriq PRISM Laboratory system without altering the three certificates or the Cantorization theorem.

## 9 Discussion and scope

The three coefficient profiles illustrate different finite mechanisms. Repeated 2's and 3's in  $a$  create a spectrum that visits every multiplicity through five. In  $b$ , the factor  $(1 + z^6)$  overlays two shifted copies of the five-term prefix spectrum, raising selected coefficients to six while avoiding four. The seven-atom witness  $c$  is more unusual: its central overlap region produces multiplicities three, four, and five while the complete spectrum avoids multiplicity two altogether.

The common Cantorization theorem separates the finite combinatorial problem from the topology of the resulting infinite achievement set. Once an integer-valued finite certificate is known, a uniquely represented tail of diameter less than one transfers the finite multiplicity range to a Cantor set without introducing collisions.

This manuscript does not determine whether  $M_5$ ,  $M_{356}$ , or  $M_{1345}$  belongs to the Cantorval class  $\mathcal{C}_v$ . It also does not revisit the interval and interval-filling classifications established or excluded in the source paper.

### Priority and review caveat

The polynomial identities are exact, but a targeted web and bibliographic search is not an exhaustive priority search. Equivalent witnesses may exist in unpublished notes, correspondence, computational data, or literature indexed under different terminology. The results should be cited as candidate classification updates until independent experts confirm both the arguments and their novelty.

## Reproducibility statement

The source package accompanying this preprint contains the complete Lua $\text{\LaTeX}$  manuscript, formula-derived cover and figures, exact verification and bounded-search scripts, recorded outputs, release metadata, checksums, and PDF quality-control reports. All proof-critical finite calculations use integer arithmetic; the tail check uses exact rational arithmetic.

## Institutional disclaimer and computational research declaration

This manuscript is an exploratory research preprint issued by the Metriq PRISM Laboratory, an interdisciplinary research laboratory of Metriq Foundation, Inc. PRISM—the Program for Research in Intelligent Systems and Methods—conducts research involving computational, mathematical, scientific, and machine-assisted methods.

The work was developed as part of PRISM's experimental research program in computationally assisted mathematics. Metriq Foundation, acting through the PRISM Laboratory, directed the research process, evaluated the resulting argument, determined the form and scope of the manuscript, and accepts responsibility for its publication as a candidate result.

Generative artificial intelligence and other computational tools were used as research instruments to support construction search, symbolic and numerical analysis, literature review, proof development, verification, and manuscript preparation. Their use does not constitute independent validation of the results presented.

This manuscript has not yet undergone independent peer review and should not be regarded as an established resolution of the stated problem unless and until its arguments have been examined and confirmed by qualified mathematicians. Independent verification, criticism, replication, and identification of errors are expressly invited.

## Creator and contributor statement

Metriq PRISM Laboratory is the organizational creator and issuing research laboratory for this candidate edition. Daniel H. Jeffery served as Research Director and corresponding contributor, with responsibility for conceptualization, project administration, supervision, resources, funding acquisition, and writing—review and editing.

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## Data and code availability

No external empirical dataset was used. The complete  $\text{Lua}\text{La}\text{T}\text{E}\text{X}$  source, figures, verification code, recorded output, review guide, and release checksums are included in the source and reproducibility package associated with this candidate edition. Computational checks verify only the claims explicitly encoded by those programs and do not independently validate unencoded infinite arguments, interpretation, or novelty.

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## Competing interests

Daniel H. Jeffery is the founder and Executive Director of Metriq Foundation, Inc. No other competing interests are declared.

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## Identifier and version history

This candidate edition, MF-PRISM-MATH-2026-06 Version 2.2, is the canonical successor to MF-MATH-2026-06 Version 2.1. The historical identifier and release remain preserved for provenance. Adoption of the laboratory-wide identifier does not reset the original publication chronology or erase superseded versions.

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